

Vishruth M V

Undergraduate, Computer Science & Engineering

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Education

B.N.M Institute of Technology , BE in Computer Science	Dec 2022 – June 2026
- GPA: 9.34/10.0 Coursework: Data Structures and Algorithms, Computer Architecture, Operating Systems, Theory of Automata and Compiler Design, AI and ML, Computational Theory	

Skills

Languages:	Python, C, SQL
Web Technologies:	HTML, CSS, JavaScript, JSP, PHP
Software Engineering:	Object-Oriented Programming, Version Control Systems, Algorithmic Design, SDLC, Agile Development - Scrum, Code Reviews, Testing, Debugging
Development Tools:	Flask, FastAPI, Docker, Git, GitHub Actions, CI/CD, RESTful APIs, Model Versioning
Databases:	MySQL, SQLite
Tools & Platforms:	Visual Studio Code, Jupyter Notebook, Google Colab, Linux, Git, GitHub
Cloud (Basics):	AWS (EC2, S3, SageMaker), GCP (Vertex AI), Azure ML
Soft Skills:	Design Thinking, Analytical Thinking, Logical Reasoning, Communication, Collaboration and Time Management

Experience

SDE 1 Intern, Quoqo Technologies Pvt Ltd.	Dec 2025 - Present
- Designed and maintained scalable RESTful backend APIs, measured by stable performance and predictable behavior under increasing production load. - Hardened backend integrations and deployment pipelines, measured by fewer post-release regressions and smoother CI/CD rollouts.	
Technical Intern - Automation & Data Systems , Alstom Transport India Limited	Aug 2025 – Dec 2025
- Developed a production-ready Excel Automation tool using Python, OpenPyXL, and VBA, reducing data processing time by 50-60%. - Built robust validation and reporting modules ensuring 100% data integrity in safety-critical railway signalling workflows across 100+ projects. - Improved tool compatibility and error-handling, reducing macros-related failures by 80% and increasing overall system adoption.	
Project Intern , Deputy Director - Bengaluru South, Dept. of Pre-University	Dec 2024 – Jun 2025
- Designed and implemented a scalable administrative software system using Python and SQL, managing data for 400+ colleges under SDLC guidelines. - Led requirement discovery and stakeholder alignment efforts by coordinating with multiple government stakeholders and officers, translating policy and operational pain points into clear technical requirements, team milestones, and committed deliverables. - Designed Test Cases and debugged system errors and optimized data retrieval functions, reducing query latency by 60%.	

Publications

- S. Geetha, S. A. P., V. D., and Vishruth M. V., “*Soft Alert Generation for Student Dropout Mitigation and Proactive Management by Machine Learning Algorithms*”, 2025 International Conference on Intelligent and Innovative Technologies in Computing, Electrical and Electronics (IITCEE), Bangalore, India, pp. 1–5, 2025. DOI: 10.1109/IITCEE64140.2025.10915221
- Raghavendra C. K., Vishruth M. V., Smaran N. G., Spandana A. P., Vijay D., “*Identifying Patterns and Trends in Campus Placement Data Using Machine Learning*”, International Journal of Education and Management Engineering (IJEME), Vol. 15, No. 1, pp. 10–24, 2025. DOI: 10.5815/ijeme.2025.01.02

Projects

AI powered Early Warning Dropout Mitigation System

- Accomplished a 35% reduction in potential student dropouts as measured by predictive model outcomes and system engagement from 1000+ users by developing a full-stack platform using Python, Flask, and SQL with Aadhaar-verified support systems aligned to NEP 2020.
- Accomplished targeted student support for 1000+ individuals as measured by 85% model accuracy in identifying dropout risks by developing ML models using academic, behavioral, and socio-economic indicators, and launching an Aadhaar-verified intervention portal.

Natural Language to SQL Generator with Automated Database Execution

- Accomplished a 92% model accuracy and 70% accessibility improvement for non-technical users as measured by testing results across diverse query samples by designing an NLP system using Python and SQLite that converts English to SQL and executes queries via a custom-built pipeline.
- Built a full pipeline (Python, SQLite) with dynamic DB creation, real-time query execution, and security validation cycles, version-controlled using Git, and enhanced through systematic debugging delivering a 5% performance gain per iteration.

Identifying Trends and Patterns in Campus Placement Data

- Analyzed multi-year campus placement datasets using Python (Pandas, NumPy, Matplotlib) to uncover trends across departments, roles, company types, and compensation levels.
- Accomplished a 30% increase in placement prediction accuracy as measured by model performance and dashboard insights by analyzing multi-year campus placement datasets using Python libraries and building predictive models with visualization dashboards.

Blood Bank Management System

- Accomplished a 40% improvement in emergency response coordination and enhanced system reliability as measured by reduced inventory retrieval time and real-time alert effectiveness by engineering a modular, full-stack Java and PHP application with MySQL integration, implementing secure authentication, CRUD operations, and automated alert systems following SDLC practices and database normalization principles.

Certificates

- **AI Business Intelligence Analyst**, Skill India / NCVET – NASSCOM (570 Hours, Level 5 – PMKVY), Issued: May 2025
- **Strategy Formulations and Data Visualization Workshop**, IIT Madras
- Secured 2nd place in the 8th National Level IEEE Project Expo 2024 for developing Proactive Dropout Mitigation for School Student Success, a predictive analytics platform, organized by IEEE Student Branch at GSSS Institute of Engineering and Technology for Women, Mysuru (May 2024)
- National-level winner, ranked 23rd among top 50 teams in a 20-hour hackathon hosted by BITS Pilani (Oct 2024), for ideating Predictive Tools for Hospital Resource Management, a data-driven healthcare efficiency solution